THE UNIVERSITY OF LAHORE

Faculty of Engineering and Technology
Department of Technology
Fall Semester 2026

Instructor's Name:	**Tahir Ilyas**	**Subject:**	**Advanced Back-End Development**
Subject Code:	**CS02338|11**	**Assignment #:**	**02**
Total Marks:	**100**	**Assigning Date:**	**October 07, 2026**
Due Date& Time:	**October 14, 2026**		

Instructions for Candidates

i. Read the instructions carefully.
ii. Write Question number and its part number clearly.
iii. Attempt all questions.

Question # 01: (CLO1, C3, PLO3) (30 Marks)

You are given a list of integers. **USE** python programming to write a program that performs the following operations using functions:

- Write a function called remove_duplicates that takes a list of integers as input and returns a new list with duplicate elements removed.
- Write a function called find_average that takes a list of integers as input and returns the average of all the elements in the list.
- Write a function called multiply_by_scalar that takes a list of integers and a scalar value as input and returns a new list where each element is multiplied by the scalar.

Question # 02: (CLO1, C3, PLO3) (30 Marks)

You are given a dictionary representing the inventory of a store. **USE** python programming to write a program that performs the following operations using loops:

- Write a loop that iterates through the dictionary and prints each item along with its quantity in the inventory.
- Write a loop that calculates and prints the total value of the inventory (quantity * price) for all items.

Question # 03: (CLO1, C3, PLO3) (40 Marks)

You are tasked with creating a class to manage a library system. The class should have the following features. **CONSTRUCT** a class called Library with the following methods:

- __init__(self, name): Initializes the library with a name.
- add_book(self, book): Adds a book to the library's collection.
- remove_book(self, book_title): Removes a book from the library's collection.
- search_book(self, keyword): Searches for a book by keyword in the library's collection and returns a list of matching titles.
- display_books(self): Displays all the books in the library's collection.
- The library's collection of books should be stored as a list of strings (book titles).